

Independent Statistical Review

BISMILLAHIR RAHMANIR RAHIM

PROJECT A PRO — A1

RUN 1 INDEPENDENT SCORING + STATISTICAL REVIEW

ROLE

You are the independent analytical/statistical reviewer.

You are NOT the original collector.

You are NOT the benchmark author.

You must not assume Codex's collection or scoring claims are correct.

INPUT

Use the attached:

A1_Run1_Post_Amendment_Full_Evidence_Package.zip

CONTROL STATE

- **Run 1 collection is complete: 24/24 definitive cases.**
- **Model under test: OpenAI gpt-5.6-luna.**
- **Reasoning setting: low.**
- **Frozen Amendment 1.1 governs parsing, scoring, provenance and review.**
- **Do NOT alter frozen references.**
- **Do NOT alter raw model responses.**
- **Do NOT alter scoring policy.**
- **Do NOT collect replacement responses.**
- **Do NOT silently harmonise disagreements.**
- **Do NOT convert selected-case results into population claims.**

PRIMARY OBJECTIVE

Independently determine:

- 1. whether Run 1 evidence is internally valid;**
- 2. what deterministic scoring can legitimately establish;**

3. what requires semantic/human review;
4. whether each model response is statistically defensible;
5. where the model fails across:
 - Evidence
 - Assumptions
 - Method
 - Reasoning
 - Uncertainty
 - Validity
 - Conclusion;
6. whether Run 1 is suitable for bounded public reporting.

=====

1. PACKAGE / PROVENANCE VERIFICATION

=====

Independently verify:

- ZIP integrity
- SHA-256 manifest
- 24 unique case IDs
- prompt hashes
- response hashes
- task provenance
- timestamps
- model identity
- reasoning setting
- run manifest
- Amendment 1.1 source hash
- no mixing with Run 0

Report any discrepancy before scoring.

=====

2. NUMERICAL RECOMPUTATION

=====

For every case with quantitative content:

Recompute the relevant quantities independently from the public case data.

Do NOT merely compare with the frozen reference.

For each quantity state:

- independent derivation
- independently computed value
- model value if present
- reference value
- agreement / disagreement
- tolerance status
- whether the model's final statistical interpretation is defensible

Keep:

NUMERICAL CORRECTNESS

separate from:

STATISTICAL VALIDITY

A valid number with an invalid conclusion must be identified explicitly.

=====

3. DETERMINISTIC SCORING REPLAY

=====

Reproduce the Amendment 1.1 deterministic scoring logic.

Verify:

- H01 requested/supplemental handling
- H02 tolerance
- H03 parsing and unit policy
- H04 NOT_APPLICABLE denominators
- duplicate-key handling
- response completeness
- parse status
- required-field status
- numeric correctness

Do not infer correctness where the parser cannot legitimately extract a value.

If Codex reported 20 applicable cases but 0 deterministically evaluated,

explain exactly why and whether that reflects:

- A. plain-prose response formatting,
- B. parser limitation,
- C. scoring-policy design,
- D. actual model failure.

Do not convert parser failure into mathematical failure.

=====

4. SEMANTIC REVIEW

=====

Review all 24 responses across the seven candidate dimensions:

- Evidence
- Assumptions
- Method
- Reasoning
- Uncertainty
- Validity
- Conclusion

Use the frozen semantic-review protocol.

For every scored dimension provide:

- run_id
- case_id
- response SHA-256
- score
- concise rationale
- reviewer identity/role
- rubric version
- adjudication status

Do NOT claim these dimensions are validated psychometric constructs.

=====

5. MATCHED-CONTRAST ANALYSIS

=====

Treat the benchmark as 12 matched contrasts, not 24 iid population samples.

Analyse paired behaviour where relevant.

Do NOT compute or imply population-level confidence intervals from the 24 selected prompts.

Report descriptive matched-case patterns only.

=====

6. FOUR PENDING STATISTICAL-ADJUDICATION CASES

=====

Independently review:

Q142
Q216
Q731
Q976

For each distinguish:

- A. model error
- B. frozen-reference risk
- C. genuine statistical ambiguity
- D. no material issue

Return one of:

MODEL ERROR
REFERENCE RISK
ADJUDICATION REQUIRED
NO ISSUE

Do not penalise the model merely for disagreeing with challenged prose if its statistical reasoning is defensible.

=====

7. FAILURE LOCALISATION

=====

Where a response is weak or wrong, identify the minimum failure point:

Evidence

- Assumptions
- Method
- Reasoning
- Uncertainty
- Validity
- Conclusion

Also state:

- highest-consequence defect
- minimum valid repair

Do not infer hidden chain of thought.
Evaluate only the observable response.

=====

8. RESULT SUMMARY

=====

Produce:

- numeric-applicability denominator
- parse-status counts
- deterministic numeric-correctness counts where legitimate
- semantic score distributions
- matched-contrast descriptive patterns
- cases requiring adjudication
- cases with correct number but invalid conclusion
- cases with invalid number but otherwise valid reasoning
- cases with uncertainty/causal-overreach defects
- strongest and weakest response categories

Avoid a single headline "accuracy" score unless methodologically justified.

=====

9. CLAIM BOUNDARIES

=====

State what the run supports and does NOT support.

Permissible examples:

- observations on these 24 specified cases

- descriptive matched-case patterns
- one-model/one-run findings
- examples of numerical/statistical failure modes

Prohibit unsupported claims such as:

- broad statistical competence
- benchmark superiority
- general population model accuracy
- stable rankings
- universal reasoning coverage
- validated seven-factor measurement
- causal localisation of internal model cognition

=====

10. OUTPUT

=====

Return:

1. independent verification report
2. deterministic replay results
3. per-case quantitative recomputation
4. per-case seven-dimension review
5. matched-contrast analysis
6. Q142/Q216/Q731/Q976 adjudication table
7. disagreement log
8. publication-safe result summary
9. exact unresolved issues
10. final recommendation

FINAL DISPOSITION:

GO FOR FINAL ADJUDICATION
 CONDITIONAL GO
 REVISE
 FAIL

Then answer:

- A. Is Run 1 admissible for final A1 adjudication?
- B. Is it admissible for bounded public reporting?
- C. What should Astra challenge independently?
- D. What exact claims are currently defensible?

Proceed independently.

Kimi K3

I'll begin by extracting and inventorying the evidence package before any analysis. ZIP integrity is confirmed (66 KB, 67 entries, all text — small enough to process fully here). Extracting and verifying the manifest next. Provenance structure confirmed. Running full cross-verification of provenance ↔ manifest ↔ files, timestamp ordering, and template reconstruction. First 12 responses read — Q195 already shows a clear numeric and verdict error (0.095238 vs the correct ≈ 0.005510 ; claim wrongly rejected). Reading the remaining 12. All 24 responses reviewed qualitatively. Now performing the independent numerical recomputation for every quantitative case, plus a systematic format analysis explaining the 0/20 parse outcome. All verification and recomputation work is complete. Here is the full independent review.

A1 Run 1 — Independent Scoring + Statistical Review

Reviewer: Kimi K3 (Abacus.AI ChatLLM Teams) — independent analytical/statistical reviewer. Not the collector, not the benchmark author. **Review date:** 2026-09-11.

Basis: package contents only; nothing taken on trust from collector narratives where re-derivation was possible.

1. Independent Verification Report (Package / Provenance)

Check	Result	Evidence
ZIP integrity	PASS	testzip() clean; container SHA-256 (my computation, external to manifest): 5c5aaa3b...d78b589
SHA-256 manifest	PASS	66/66 listed files rehashed and match; no missing, no extra files (manifest excludes itself, correctly)
24 unique case IDs	PASS	24/24 unique; provenance order = manifest execution order exactly
Prompt hashes	PASS	All 24 prompts/Q*.txt rehashed = provenance = manifest = deterministic_checks.json
Response hashes	PASS	All 24 case_raw_responses/Q*.txt rehashed = provenance = manifest = queue records
Template reconstruction	PASS	All 24 exact_prompt strings rebuilt byte-identically from user_prompt_template.txt + exact_case_payload (compact JSON serialization)
Raw JSONL	PASS	Whole-file hash = manifest raw_response_sha256; per-record response text byte-identical to per-case files
Task provenance	PASS (internal)	24 unique task IDs, distinct rollout filenames, projectless directory names, all turns complete; task-ID prefixes corroborate two collection sessions (01a08e6b-6d vs 01a09250 for Q903/Q928/Q976)
Timestamps	PASS, with observation	All response timestamps within the declared window and monotone in execution order. Observation: the "run" spans 18.16 h with a ~17.7 h gap between Q854 and Q903 — 21 cases in a ~34 min morning window, 3 cases (Q903, Q928, Q976) in an evening window. Not a violation of fresh-task isolation, but "one run" = two sessions and should be disclosed. Rollout filenames are offset +1 h from UTC (local-time naming, consistent with BST); not a discrepancy.
Model identity / reasoning setting	PASS (internal)	gpt-5.6-luna / OpenAI / reasoning low consistent across manifest, config, provenance, and all 24 JSONL records. Platform-side identity is not independently verifiable from the package; it rests on the collector's recovered task histories.
Zero tool/browser/memory invocations	PASS (recorded)	All counts 0 in all 24 provenance records; unverifiable beyond the records
Retries/refusals/truncations	PASS	0/0/0/0, corroborated by provenance statuses

Check	Result	Evidence
Amendment 1.1 source hash	CONSISTENT, NOT RE-VERIFIABLE	01e7092d...602d75 recorded identically in manifest, execution log, and checks (expected = actual). The Amendment ZIP itself is not packaged — I could not rehash it, read the frozen parser, the tolerance rule, the reference values, or the rubric anchors. This bounds §3 and §4 below.
Run 0 mixing	NO INTERNAL EVIDENCE; PARTIALLY ASSERTIVE	Recorded overlap count = 0 with empty case list; run0_file_used_as_response_source=false. However, the Run 0 package was unavailable for rehash (correctly labelled "historical metadata only"), so the overlap check was executed by the collector against data I cannot inspect. Separation is an internal-consistency pass, not an independent verification.
case_set_sha256	NOT REPRODUCIBLE	Derivation method unspecified; two obvious reconstructions fail. Not contradicted; unverifiable.
Duplicate artifacts	Noted	deterministic_scoring.jsonl ≡ derived/derived_parsing_and_scores.jsonl and the two summaries are byte-identical copies — mirrors, not conflicts.

Discrepancies found before scoring: none affecting integrity. Four items are unverifiable from package contents (Amendment source hash, Run 0 separation, case_set_sha256, platform-side model identity) and are carried as open items in §9.

2. Deterministic Replay Results (Amendment 1.1 logic)

The frozen parser source is not packaged, so this is a **consistency replay** against the recorded outputs and the stated parsing rule ("valid JSON, supported JSON-like syntax, or an unambiguous single number"), not a re-execution of frozen code.

Component	Replay verdict
H01 requested/supplemental separation	Consistent. Records score only required_numeric_fields; supplemental_reference_fields_excluded empty everywhere (vacuously, since nothing parsed).
H02 tolerance	NOT REPLAYABLE. No reference values exist anywhere in the package; the tolerance window cannot be checked. Structural finding: the packaged derived layer contains <i>no</i> reference values, so no one outside the frozen repository can reproduce a correctness verdict from this package alone.
H03 parsing/unit policy	Consistent. I classified all 24 responses: none is whole-response valid JSON or a single bare number. 15 use labelled key: value prose lines, 4 use LaTeX display math, 2 use fenced code blocks, Q768 leads with a complete valid JSON object followed by prose. Every recorded parse_detail ("not valid JSON...") is accurate. Unit policy untriggered (no parsed values to unit-check).
H04 NOT_APPLICABLE denominators	Consistent. Exactly the four cases whose task text says "No numerical result is requested" (Q187, Q542, Q639, Q976) are NOT_APPLICABLE; 20 APPLICABLE; denominators in the summary follow correctly.
Duplicate-key handling	Not exercisable. No response reaches the parse stage where duplicate keys could arise. Branch unverified.
Response completeness	PASS. 24/24 present, non-empty, hash-verified.
Parse status	24/24 UNPARSED — reproduced as consistent with the stated rule.
Required-field status	20 FAIL / 4 NOT_APPLICABLE — follows deterministically from UNPARSED.
Numeric correctness	0 evaluated, 0 passed — follows deterministically.

Why 20 applicable but 0 evaluated — adjudication of cause. This is **A + C**, not **B**, not **D**:

- **A (plain-prose formatting):** the frozen neutral instruction demands a "self-contained answer" preserving field names but never requires JSON or machine-readable output. The model answered in labelled prose — compliant with the instruction it was given.
- **C (scoring-policy design):** the parser deliberately accepts only whole-response JSON / JSON-like / single number, and by policy converts everything else to NOT_EVALUATED rather than wrong. The instruction layer and the parser layer are mismatched by design.
- **Not B (parser limitation in the malfunction sense):** every recorded parse decision is *correct under the stated rule*. Q768 is the edge case — its first line is complete valid JSON with all four requested fields correct; only trailing prose defeats whole-response parsing. A marginally extracting parser would have scored Q768 and Q854 (fenced JSON block, values correct). That is a policy choice, not a bug.

- **Not D (model failure):** my recomputation (§3) shows 19/20 applicable cases returned every requested value correctly. **The parser failure must not be converted into a mathematical-failure claim — and equally, the "0 passes" must not be reported as 0% accuracy.**
-

3. Per-Case Quantitative Recomputation (independent, from public case data)

Reference values are not packaged — the reference column is therefore "n/p" throughout; my independently computed values are the primary check. Tolerance status: not replayable (no reference, no tolerance rule); all agreements below are exact to the model's stated precision (≥ 6 d.p. rounding verified).

Case	Quantity	Independent derivation	Independent value	Model value	Agree	Interpretation defensible
Q142	slope / intercept / fitted@8	$\Sigma(x-\bar{x})(y-\bar{y}) / \Sigma(x-\bar{x})^2 = 6/10; \bar{y}-b \cdot \bar{x}; \hat{y}(8)$	0.6 / 2.2 / 7.0	0.600000 / 2.200000 / 7.000000	✓	Yes — $x=8$ outside [1,5] is extrapolation; fitted $\neq$ guaranteed observation
Q195	fault_given_negative	$(1-\text{sens}) \cdot p / [(1-\text{sens}) \cdot p + \text{spec} \cdot (1-p)] = 0.005/0.9075$	0.005509642	0.095238095	X (17.29x high)	No — claim (" $<1\%$ ") is true ; model said false. Denominator used $p \cdot \text{spec} = 0.0475$ instead of $(1-p) \cdot \text{spec} = 0.9025$ (mechanism confirmed: $0.005/0.0525 = \text{model's value exactly}$)
Q216	risk_difference	$36/60 - 24/60$	0.2	0.200000	✓	Yes — random assignment stipulated; estimation $\neq$ proof explicitly distinguished
Q251	recall / pred_pos_count	$0/30; \text{TP}+\text{FP}$	0 / 0	0.000000 / 0	✓	Yes — precision 0/0 undefined; matches analyst
Q268	adjusted_p	$\min(1, 1 \times 0.012)$	0.012	0.012000	✓	Yes — reject at FWER 0.05
Q317	ppv	$0.024/(0.024+0.0097)$	0.712166172	0.712166	✓	Yes — $0.712 > 0.5$, claim true

Case	Quantity	Independent derivation	Independent value	Model value	Agree	Interpretation defensible
Q361	selected_slope / intercept / rss	$\Sigma xy / \Sigma x^2 = 66/55; 0; \Sigma (y-1.2x)^2$	1.2 / 0 / 6.8	1.200000 / 0.000000 / 6.800000	✓	Verdict yes, support no — minimization-claim verdict (false) is correct, but the model's <i>supplemental</i> free-fit values (slope 0.400000, intercept 2.800000, RSS 2.800000) are all wrong: true free OLS is 0.6 / 2.2 / 2.4. Ordering (2.8 < 6.8) luckily preserves the conclusion (2.4 < 6.8)
Q423	cost_a/b, acc_a/b	18+2; 2+8; 80/100; 90/100	20 / 10 / 0.8 / 0.9	as computed	✓	Yes — B lower-cost, recommendation correct
Q459	nominal_lower/upper	12 ± 1.959963984 540054×6/ √36	10.04003601 5459946 / 13.95996398 4540054	10.04003601 5460 / 13.95996398 4540	✓	Yes — independent units + known σ : z-interval valid, no correction needed
Q473	a_pooled / b_pooled	96/140; 130/170	0.685714286 / 0.764705882	0.685714 / 0.764706	✓	Yes — claim (A>B pooled) false; observational ⇒ no causal claim
Q586	slope / intercept / fitted@3	as Q142; $\hat{y}(3)$	0.6 / 2.2 / 4.0	0.600000 / 2.200000 / 4.000000	✓	Yes — x=3 interpolation claim correct
Q624	fault_given_negative	0.06/(0.06+0.38)	0.136363636	0.136363636	✓	Yes — claim false
Q694	recall / pred_pos_count	as Q251	0 / 0	0.000000 / 0	✓	Yes — "precision is zero" correctly rejected as invalid without a declared convention

Case	Quantity	Independent derivation	Independent value	Model value	Agree	Interpretation defensible
Q731	nominal_lower/upper (+ correction)	as Q459; eff. $n=1 \Rightarrow 12 \pm 1.959963984540054 \times 6$	10.040036 / 13.959964 (corr: 0.240216 / 23.759784)	identical	✓	Yes — perfectly correlated repeats of one unit $\Rightarrow$ effective $n=1$; model's corrected interval is the defensible repair
Q768	cost_a/b, acc_a/b	18+20; 2+80	38 / 82 / 0.8 / 0.9	38 / 82 / 0.800000 / 0.900000	✓	Yes — recommendation of B correctly rejected
Q795	selected_slope / intercept / rss	free OLS	0.6 / 2.2 / 2.4	0.600000 / 2.200000 / 2.400000	✓	Yes — claim true
Q842	ppv	0.024/(0.024+0.097)	0.198347107	0.198347	✓	Yes — claim false
Q854	risk_difference	0.6 – 0.4	0.2	0.200000	✓	Yes — self-selection $\Rightarrow$ not design-justified; estimation $\neq$ proof distinguished
Q903	adjusted_p	min(1, 20×0.012); threshold 0.0025	0.24	0.240000	✓	Yes — no rejection
Q928	a_pooled / b_pooled	as Q473	0.685714 / 0.764706	identical	✓	Yes — within-group claim true ($0.84 > 0.80$; $0.60 > 0.50$); Simpson structure explained; causal caution present

Numerical correctness vs statistical validity, separated: requested-field numeric correctness is 19/20 applicable cases (sole failure: Q195). Statistical-validity failure: Q195 (inverted verdict). Q361 is the single instance of *invalid supplemental numbers with a valid requested output and valid conclusion*; there is **no** instance of a correct number carrying an invalid conclusion.

4. Per-Case Seven-Dimension Semantic Review

Material caveat: the frozen Amendment 1.1 rubric anchors/scale are not in the package (queue records contain only the rubric name, no anchors). I therefore apply a **declared 3-level reviewer scale — PASS / MINOR DEFECT / FAIL —** and these scores are my independent judgments, *not* a reconstruction of frozen rubric scores, and *not* validated psychometric constructs. Fixed fields for all 168 decisions: run_id a1-run1-post-amendment-gpt-5.6-luna-20260911T030318Z-20260911T211259Z; reviewer "Kimi K3, independent analytical/statistical reviewer"; rubric version "reviewer-declared 3-level scale (frozen A1 Amendment 1.1 anchors unavailable in package)"; adjudication status UNADJUDICATED pending Astra; review date 2026-09-11. Response SHA-256 prefixes below key each row (full hashes verified in §1).

Case (hash prefix)	Evi	Ass	Met	Rea	Unc	Val
Q142 092fbf6b	P	P	P	P	P	P
Q187 3a140937	P	P	P	P	P	P
Q195 cd6b9c87	P	P	F	F	P	F
Q216 691db58d	P	P	P	P	P	P
Q251 594f8a18	P	P	P	P	P	P
Q268 e71e1303	P	P	P	P	P	P
Q317 0dd165b9	P	P	P	P	P	P
Q361 2b50ad76	M	P	P	M	P	P
Q423 31afa61c	P	P	P	P	P	P
Q459 e93aaae5	P	P	P	P	P	P
Q473 6eae1e98	P	P	P	P	P	P
Q542 559175a9	P	P	P	P	P	P
Q586 48b1c537	P	P	P	P	P	P
Q624 8c734c80	P	P	P	P	P	P
Q639 7d6c05e2	P	P	P	P	P	P
Q694 78a14e1d	P	P	P	P	P	P
Q731 fab28ecc	P	P	P	P	P	P
Q768 db23b79b	P	P	P	P	P	P
Q795 a89c0dbd	P	P	P	P	P	P
Q842 3bb7f846	P	P	P	P	P	P
Q854 a3c5357e	P	P	P	P	P	P
Q903 cf73a850	P	P	P	P	P	P
Q928 58e0c5c0	P	P	P	P	P	P

Case (hash prefix)	Evi	Ass	Met	Rea	Unc	Val
Q976 f2239840	P	P	M	P	P	P

+1 more column not shown: Con

Distribution: 161 PASS / 3 MINOR / 4 FAIL across 168 decisions. Rationales for all non-PASS cells (everything else passed on the strength of the recomputation and verdicts in §3):

- **Q195 Method — FAIL:** Bayes denominator assembled with prevalence×specificity (0.0475) in place of (1−prevalence)×specificity (0.9025); mechanism confirmed by exact digit match. **Reasoning — FAIL:** derivation internally consistent but propagates the method error to a 17.29× overestimate. **Validity — FAIL:** the challenged claim (posterior < 1%) is true (0.551%); the model declared it false. **Conclusion — FAIL:** wrong value and inverted verdict. Evidence/Assumptions/Uncertainty pass: correct inputs selected, stipulated parameters respected, exact-parameter framing appropriate in form.
- **Q361 Evidence — MINOR:** asserted comparator (free-fit) values 0.4/2.8/2.8 are inconsistent with the supplied data. **Reasoning — MINOR:** comparator recomputation wrong; conclusion survived only because the erroneous RSS (2.8) preserves the true ordering ($2.4 < 6.8$). Method/Validity/Conclusion pass: the right comparison was chosen, the verdict is correct, and all three *requested* fields are exact.
- **Q976 Method — MINOR:** the alternative repair "delay scoring until 10:05" relaxes the stated requirement (score at 10:00) rather than meeting it; the primary repair (exclude the feature) is valid and minimal, so the defect is confined to one disjunct.

Notable PASS nuances: Q542's "repair" is preventive hardening, correctly not framed as a defect fix; Q216/Q854 both explicitly distinguish estimation from proof of a nonzero effect; Q142/Q586 both distinguish fitted values from guaranteed observations; Q473/Q928 both refuse causal claims from observational data. **No causal-overreach or unwarranted-certainty defect occurs anywhere in the 24 responses** — a genuine strength of this run.

5. Matched-Contrast Analysis (12 pairs, descriptive only)

The 24 cases are 12 designed contrasts, not iid draws. No confidence intervals, no population claims.

Pair	Contrast manipulated	Model behaviour
Q142 / Q586	x=8 extrapolation vs x=3 interpolation	Correct discrimination both arms
Q187 / Q639	imputer fit on all rows vs training only	Correct both arms (leakage found; clean pipeline approved)
Q195 / Q624	prevalence 0.05 (claim true) vs 0.60 (claim false)	Failed discrimination: wrong value + verdict at low prevalence; correct at high prevalence. Error mechanism invisible in the Q624 arm — prevalence-side denominator term
Q216 / Q854	random assignment vs participant choice	Correct both arms; clean causal/design separation
Q251 / Q694	analyst says "undefined" vs says "zero"	Correct both arms (affirmed undefined; rejected zero)
Q268 / Q903	m=1 vs m=20 Bonferroni	Correct both arms (0.012 reject; 0.24 no-reject)
Q317 / Q842	FPR 0.01 (PPV 0.71) vs 0.10 (PPV 0.20)	Correct both arms
Q361 / Q795	constrained vs free intercept minimization claim	Verdicts correct both arms; supplemental numeric corruption in Q361 arm only
Q423 / Q768	FN cost 1 vs 10 (flips optimal model)	Correct both arms — cost-sensitivity tracked
Q459 / Q731	36 independent units vs 1 unit ×36 correlated repeats	Correct both arms; valid repair (effective n=1) in Q731
Q473 / Q928	pooled claim (false) vs within-group claim (true) — Simpson structure	Correct both arms
Q542 / Q976	snapshot at 09:55 (available) vs 10:05 (lookahead)	Correct both arms

Descriptive pattern: 11/12 contrasts show correct verdict-level discrimination in both arms. The single discrimination failure (Q195/Q624) and the single support-level corruption (Q361/Q795) are both numeric-computation defects under verbatim-parameter arithmetic, not conceptual failures — the same model articulates extrapolation, leakage, confounding, multiplicity, Simpson's paradox, effective sample size, and lookahead bias correctly in prose. Failure concentrates in *executing* Bayes arithmetic in one parameter regime.

6. Adjudication Table — Q142 / Q216 / Q731 / Q976

Case	A. Model error?	B. Reference risk?	C. Genuine ambiguity?	D. Material issue?	Return
Q142	None — values exact (0.6/2.2/7.0); extrapolation correctly identified; fitted $\neq$ guaranteed stated	Cannot test: reference values/text not packaged	None — "estimated mean response" is standard for an OLS fitted value	—	NO ISSUE
Q216	None — RD=0.2 exact; design-justified estimation under stipulated randomization + SUTVA + complete follow-up is textbook-defensible; estimation/proof distinction explicit	Cannot test from package; if the frozen reference prose demands a different verdict wording, that is a reference-side matter, not a model defect	None material	—	NO ISSUE
Q731	None — nominal bounds exact; invalidity under perfect correlation correct; effective $n=1$ correction ($12 \pm 1.96\sigma$) is the defensible repair given unit-level σ	Cannot test: if the frozen reference specified a different "required correction", the package does not show it	Minor in principle (σ interpretation), but scenario stipulates "population of unit values", resolving it	—	NO ISSUE
Q976	None — lookahead correctly identified; exclusion repair valid; secondary "delay to 10:05" repair is requirement-relaxing (scored MINOR, §4)	Cannot test from package	Minor: whether delayed scoring is operationally acceptable is requirement-dependent	None material	NO ISSUE

All four flagged responses are statistically defensible as written. Nothing in the package shows what provoked the flags; on the observable evidence, none contains a model error.

7. Disagreement Log (this review vs collector artefacts)

1. **Flag sensitivity.** The package flags Q142/Q216/Q731/Q976 (all defensible — §6) and does **not** flag Q195 (the one clear model failure) or Q361 (supplemental numeric corruption). With deterministic scoring vacuous and semantic review unresolved, no package-internal mechanism could have detected them — the

flags are evidently prose-challenge-driven. Net: flag set sensitivity to actual numeric failure in this run is 0/2.

2. **Collector disposition "CONDITIONAL PASS"** rested on unresolved semantic review. This review resolves that layer (161/3/4); the disposition is superseded by §10.
3. **No disagreement** on any integrity/provenance claim — all re-derivable claims verified.
4. **FINAL_REPORT's "0 numeric passes / 0 evaluated"** is accurate and appropriately non-committal; it must be preserved verbatim in any public reporting to prevent a "0% accuracy" misreading.

Failure localisation (minimum failure point, per §7 of the brief)

- **Q195:** minimum failure point = **Method** (Bayes denominator construction) → propagates through Reasoning → Validity → Conclusion. Evidence and Assumptions intact. Highest-consequence defect: inversion of a true safety-relevant claim with a $17.29\times$ overestimate of posterior fault probability. Minimum valid repair: replace the denominator's true-negative term with $(1-0.05)\times 0.95 = 0.9025$, giving $0.005510 < 0.01 \Rightarrow$ claim true.
 - **Q361:** minimum failure point = **Reasoning** (recomputation of the free-fit comparator), not Method (comparing against free OLS is the right method). Highest-consequence defect: none material — requested outputs and verdict survive; risk is credibility if the supporting numbers are quoted. Minimum valid repair: substitute 0.6/2.2/2.4 for 0.4/2.8/2.8.
 - **Q976:** minimum failure point = **Method** (one disjunct of the repair relaxes the requirement). Minimum valid repair: retain exclusion; reframe delayed scoring as a requirement change, not a repair.
 - No hidden chain-of-thought is inferred anywhere; only observable response text is evaluated.
-

8. Publication-Safe Result Summary

- **Numeric-applicability denominator:** 20 applicable / 4 not applicable (Q187, Q542, Q639, Q976); response completeness 24/24.
- **Parse-status counts:** 24/24 UNPARSED under the frozen narrow rule; 0/20 applicable cases deterministically evaluated; deterministic numeric-correctness counts: **none legitimately exist** (denominator 0). Any "score" derived from this layer would be fabrication.

- **Independent recomputation (this review, not the frozen pipeline):** 19/20 applicable cases returned all requested values exactly (≥ 6 d.p.); 1/20 (Q195) wrong by $17.29\times$ with inverted verdict; 1 case (Q361) has correct requested values with corrupted supplemental values.
- **Semantic score distribution (reviewer-declared 3-level scale):** 161 PASS / 3 MINOR / 4 FAIL of 168; all FAILs concentrated in Q195.
- **Matched-contrast patterns:** 11/12 contrasts correctly discriminated at verdict level in both arms (descriptive only).
- **Cases requiring adjudication:** none of the four flagged cases requires corrective adjudication; Q195 and Q361 require adjudication *entry* despite being unflagged.
- **Correct number / invalid conclusion:** none. **Invalid number / otherwise valid reasoning:** Q361 (partial — supplemental only). **Uncertainty/causal-overreach defects:** none in 24 responses.
- **Strongest categories:** qualitative governance-style cases (leakage, lookahead, confounding, multiplicity, Simpson) — 12/12 verdicts correct including both arms of every conceptual contrast. **Weakest category:** verbatim multi-term probability arithmetic (Q195) and unstated-supplemental recomputation (Q361).
- **No single headline accuracy figure is methodologically justified:** the deterministic layer is vacuous by design, the semantic scale is reviewer-declared, and the 24 cases are selected contrasts, not a sample.

9. Exact Unresolved Issues

1. Amendment 1.1 source ZIP (hash 01e7092d...602d75) not packaged $\Rightarrow$ frozen parser, H02 tolerance, reference values, and rubric anchors not replayable/verifiable; my semantic scale is a declared substitute, not the frozen rubric.
2. Run 0 separation verified only to internal-consistency level; Run 0 package (42034c88...696ee) unavailable for fresh rehash.
3. case_set_sha256 derivation method unspecified; not reproducible.
4. Platform-side truth of model identity, reasoning setting, and zero-tool claims rests on collector-recovered task histories; not independently auditable from the package.
5. Sampling parameters (temperature, top_p, max tokens) and exact system/developer prompts UNAVAILABLE $\Rightarrow$ exact re-generation is impossible by design; only the controlled design is reproducible.

6. Two-session composition of the run window (~17.7 h gap before Q903) is observed, not explained; no evidence it affected isolation.
7. Origin of the four adjudication flags is undocumented in the package.

10. Final Recommendation

FINAL DISPOSITION: GO FOR FINAL ADJUDICATION — conditioned on carrying §9 as open items.

A. Admissible for final A1 adjudication? Yes. Integrity, provenance, isolation records, and template reconstruction all verify; the deterministic layer is honestly vacuous; the semantic layer is now populated (pending Astra's rubric-anchored re-scoring if the frozen anchors differ from mine); all four pending cases are cleared as defensible; two unflagged defects (Q195, Q361) are now localised and adjudication-ready.

B. Admissible for bounded public reporting? Yes, conditionally — only with: the deterministic layer reported as "0/20 evaluated by design of the narrow parser", never as accuracy; the Q195 failure and Q361 supplemental corruption disclosed; the two-session collection window disclosed; the unverifiable items (§9) disclosed; and claim language confined to §8.

C. What should Astra challenge independently? (i) Obtain and rehash the Amendment 1.1 ZIP and re-run the frozen processor to test H02 tolerances, reference values, and the duplicate-key branch — none is replayable from this package; (ii) interrogate the flag-generation process — why were four defensible responses flagged while Q195/Q361 escaped (flag sensitivity 0/2 on actual numeric defects); (iii) verify Run 0 separation externally rather than from internal assertions; (iv) obtain the case_set_sha256 derivation; (v) resolve the instruction/parser mismatch — either require machine-readable output in the neutral instruction or extend the parser, otherwise Run 2 will reproduce a vacuous deterministic layer; (vi) confirm platform-side that sampling controls were genuinely unexposed (this caps exact reproducibility).

D. Exactly defensible claims now: on these 24 specified cases, for this one model, this one run, this reasoning setting: integrity and provenance of the evidence package are fully hash-verified (66/66); 19/20 numerically applicable cases returned all requested values correctly to ≥6 d.p. against independent recomputation; 23/24 final verdicts are statistically defensible (sole exception Q195); 11/12 matched contrasts show correct two-arm verdict discrimination; zero instances of causal overreach or unjustified certainty appear in any response; and two specific numeric failure modes are documented (Bayes denominator error at low prevalence; supplemental comparator corruption with intact conclusion). Nothing broader — no competence, ranking, coverage, or validated-measurement claim — is supported.
